Pratirup · system architecture (as built)

Glucose digital twin for type 2 diabetes care teams · research prototype · Team Aflatoon, Scaler School of Technology

Static / historical stream

ShanghaiT2DM clinical record (real, de-identified, CC BY 4.0): age, sex, BMI, diabetes duration, HbA1c, C-peptide, lipids, eGFR, complications, comorbidities, glucose-lowering drugs. Genetic markers: not recorded (shown as absent).

`pratirup/data/shanghai.py`

Dynamic stream

Real CGM, every 15 min (Libre H), from held-out patients; insulin doses, oral agents, pump rates and meals as timed logs. Replayed on a demo clock, or streamed reading-by-reading into the live twin (device simulator or POST /api/live/{id}/readings).

`pratirup/data/vocab.py`

External validation (offline)

CGMacros, PhysioNet (CC BY-NC-SA 4.0): 45 US adults, healthy / prediabetes / T2D; Libre Pro + Fitbit HR and METs. Frozen-model transfer test and wearable experiment. Not shipped.

`pratirup/evaluation/external.py`

Normalise and validate

FHIR R4 bundle per record (Patient, Condition, Observation, MedicationStatement); 15-min grid; range and order checks; free-text drug and diet parsing; FHIR dataAbsentReason for genetics.

`pratirup/data/fhir.py`

Causal features

Built only from data at or before the forecast time (leakage test). CGM lags, rates, Kalman state; meals, insulin on board, oral agents; clinical record; personal time-of-day profile.

`pratirup/features.py`

clinical context

feature row

Digital twin engine (per patient)

Patient state

clinical context · Kalman level, rate, uncertainty
event context · personal profile · model trust
data quality

Forecaster

LightGBM
+30 / +60 / +120 min
conformal 80% interval

Event risk

calibrated P(>180, >250, <70 within 60 min)
alert: dev-set threshold

Self-correction

EWM of this patient's resolved forecast errors

Abstention

< 8 readings in 3 h, or last reading > 30 min old

residuals

Forecast ledger

every forecast stored, scored at its target time

`pratirup/twin/engine.py`

state, forecasts, alerts

Persistence and API

SQLite: recording, observation, event, twin_state, forecast (ledger)

FastAPI: /worklist, /patients/{id}/twin, /ledger, /sensitivity, /live/*, /evaluation, /meta

Docker: model + open data + results; no secrets; offline

`pratirup/api/`

JSON

Clinician web app

React + TypeScript
Worklist (priority by consensus targets + predicted risk) -> Patient twin (CGM, forecast band, past forecasts vs reality, drivers) -> Ledger -> What-if -> FHIR record; Live stream; Evidence (all metrics read from results/benchmark.json)

`web/`

benchmark.json

Evaluation pipeline (offline)

5-fold patient-grouped CV on 80 dev patients; 20 test patients scored once (lock file); baselines, ablations, GRU, robustness; CGMacros external test

What-if (model sensitivity)

Change one logged input (add or remove a meal / insulin), rerun the same feature builder and model, compare. Associational, not causal; out-of-range inputs flagged. No physiological simulator is claimed.

`POST /api/patients/{id}/sensitivity`

what-if request

- Real research data
- Model inference / twin state
- Model sensitivity (hypothetical)
- Clinician interaction
- Storage / processing

Dashed border: offline only, not part of the running app.